

Exam Date & Time: 16-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Algorithmic Graph Theory [MAT_4341]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) The transpose of a directed graph $G = (V, E)$ is the graph $G^T = (V, E^T)$ where $E^T = \{(v, u) : (u, v) \text{ is in } E\}$. Describe an efficient algorithm for computing G^T from G when the adjacency matrix representation of G is given. (3)
 - A)
 - B) Design an algorithm to find diameter of a tree T . Analyse the running time of your algorithm. (3)
 - C) Write an algorithm to check if a given graph is bipartite. Analyse the time complexity of your algorithm. (4)
- 2) Suppose we define a different kind of graph $G = (V, E)$ where we have positive weights on the vertices and not on the edges. Given a starting vertex v find the lowest-cost path from v to every other vertex. The cost of a path is the sum of the weights of the vertices on the path. (3)

Modify Dijkstra's algorithm to find shortest path from v to every other vertex.

 - A)
 - B) Write the steps of Bellman-Ford algorithm and analyse its time complexity. (3)
 - C) Write down the steps of Kruskal's algorithm to find minimum spanning tree in a weighted undirected graph. Analyse its time complexity. (4)
- 3) (I) Prove that a directed graph is acyclic if and only if a depth search of G yields no back edge. (4)

(II) Let T be a tree with maximum degree k . Prove that T has at least k vertices of degree 1. (4)

 - A)
 - B) Prove that Hamiltonian Cycle problem in a graph is NP-Complete. (3)
 - C) Write an algorithm to check whether a given graph is Eulerian. Analyse its time complexity. (3)
- 4) Let $G = (V, E)$ be a planar graph with n vertices and m edges. Prove that $m \leq 4n - 6$ (3)
 - A)

- B) What is chromatic number of a graph? State Brooks' Theorem. (3)
- C) Prove that a complete bipartite graph contains a perfect matching if and only if it contains equal number of vertices in both the sides of the bipartition. (4)
- 5) (I) What is the chromatic index of a complete graph having n vertices?
 (II) Prove that for any bipartite graph, the number of edges in a maximum matching is equal to the number of vertices in a minimum vertex cover. (4)
- A)
- B) Prove that the vertex cover problem is NP-Complete. (3)
- C) What is an approximation algorithm? Discuss the advantages and disadvantages of approximation algorithms. (3)

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Event Name	Subject
058-END SEMSTER-ODD-2025	MAT_4341:Algorithmic Graph Theory

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	14
3	3	CO 3	6
4	4	CO 4	14
5	5	CO 5	6
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	MAT_4341:Algorithmic Graph Theory

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	11	35(70%)
4	Analysis	4	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)